

Proof. Construction of a non-negative quadratic. Define for any real parameter t

$$f(t) = \sum_{i=1}^n (a_i - t b_i)^2.$$

Each summand is a square, therefore

$$f(t) \geq 0 \quad \text{for all } t \in \mathbb{R}. \quad (1)$$

Expansion of $f(t)$. Using $(x - y)^2 = x^2 - 2xy + y^2$,

$$(a_i - t b_i)^2 = a_i^2 - 2t a_i b_i + t^2 b_i^2.$$

Summing over i gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= At^2 + Bt + C, \end{aligned}$$

where

$$A = \sum_{i=1}^n b_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2. \quad (2)$$

Applying the discriminant condition. Since the quadratic $At^2 + Bt + C$ is non-negative for every real t (by (1)), we have

$$A \geq 0 \quad \text{and} \quad B^2 - 4AC \leq 0. \quad (3)$$

The first inequality is obvious because A is a sum of squares.

Evaluating the discriminant. Substituting (2) into $B^2 - 4AC$ yields

$$\begin{aligned} B^2 - 4AC &= (-2 \sum a_i b_i)^2 - 4 \left(\sum b_i^2 \right) \left(\sum a_i^2 \right) \\ &= 4 \left[\left(\sum a_i b_i \right)^2 - \left(\sum a_i^2 \right) \left(\sum b_i^2 \right) \right]. \end{aligned}$$

Hence (3) is equivalent to

$$\left(\sum a_i b_i \right)^2 \leq \left(\sum a_i^2 \right) \left(\sum b_i^2 \right). \quad (4)$$

This is the Cauchy–Schwarz inequality.

Degenerate case $\sum b_i^2 = 0$. If $A = 0$ then each $b_i = 0$. Consequently

$$\sum a_i b_i = 0, \quad \left(\sum a_i^2 \right) \left(\sum b_i^2 \right) = 0,$$

and (4) holds with equality.

Equality condition. Equality in (4) occurs exactly when the discriminant in (3) is zero, i.e. $B^2 - 4AC = 0$. In this case the non-negative quadratic must be a perfect square:

$$f(t) = A(t - t_0)^2$$

for some real t_0 . Evaluating at $t = t_0$ gives $f(t_0) = 0$, and because $f(t_0)$ is a sum of squares, each summand vanishes:

$$a_i - t_0 b_i = 0 \quad \text{for every } i.$$

Thus the vectors are proportional,

$$(a_1, \dots, a_n) = \lambda (b_1, \dots, b_n) \quad \text{with } \lambda = t_0 \in \mathbb{R},$$

i.e. they are linearly dependent. If all $b_i = 0$ (the degenerate situation), the proportionality condition is trivially satisfied and equality also holds. $\square$